Addie McCurdy

University of Colorado Boulder
Department of Applied Mathematics

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Education

Ph.D. in Applied Mathematics **Expected Summer 2027**
University of Colorado Boulder GPA: 3.974
Advisor: Professor Emily Jensen
Thesis title: "Spatial properties of optimal controllers over Sobolev spaces"

B.A.s in Pure Mathematics and Economics **May 2022**
University of St. Thomas - Minnesota GPA: 3.99
Honors: Summa Cum Laude, Aquinas Honors Scholar

Research Experience

Graduate Research Assistant (May 2024 – present)
University of Colorado Boulder
Mentor: Professor Emily Jensen
Focus: efficient optimal control for systems governed by partial differential equations

Graduate Research Assistant (Spring 2024)
University of Colorado Boulder
Mentor: Professor François Meyer
Focus: machine learning methods on network valued random variables

Research in Industrial Projects for Students (RIPS) (Summer 2021)
University of California, Los Angeles
Mentors: Dr. Robert Martin and Dr. Daniel Eckhardt - Air Force Research Laboratoy
Focus: deconvolution of dynamically coupled noisy signals, with application to Hall Effect Thruster data

Undergraduate Research Assistant (Feb 2019 – May 2022)
University of St. Thomas - Minnesota
Mentor: Professor Eric Rawdon
Focus: analysis and visualization of data for projects concerning new models and biological applications of mathematical knots

Honors and Awards

Sheryl R. Young Memorial Fellowship (2024 – 2025)

Research Visits

University of Michigan (Sep 2026)
Visited Professor Peter Seiler, Department of Electrical Engineering and Computer Science
Focus: integral quadratic constraints for robust control of systems governed by partial differential equations

[1] Updated November 10, 2026

Publications and Preprints

Peer-reviewed conference publications:

3. **Conditions for Complete Decentralization of the Linear Quadratic Regulator**
(**A. McCurdy**, I. Collins, E. Jensen). *27th International Symposium on Mathematical Theory of Networks and Systems*. pp. 139-144, 2026

2. **A System Level Approach to LQR Control of the Diffusion Equation**
(**A. McCurdy**, A. Gusty, E. Jensen). *American Control Conference* . pp. 4723-4730, 2026

1. **Complete Decentralization of Linear Quadratic Gaussian Control for the Discrete Wave Equation**
(**A. McCurdy**, E. Jensen). *IEEE 64th Conference on Decision and Control*. pp. 6162-6168, 2025

Under revision:

1. **Complete Decentralization of Wave Equation Control on a Compact Domain**
(**A. McCurdy**, E. Jensen). *under review at SIAM Journal on Control and Optimization*. arXiv:2611.13449

Talks and Presentations

6. *Conditions for Complete Decentralization of the Linear Quadratic Regulator*. (poster). SIAM Student Mini-Symposium in Applied Mathematics, Ann Arbor, Michigan, August 2026.

5. *Conditions for Complete Decentralization of the Linear Quadratic Regulator*, 27th International Symposium on Mathematical Theory of Networks and Systems, Waterloo, Canada, August 2026.

4. *A System Level Approach to LQR Control of the Diffusion Equation*, American Control Conference, New Orleans, Louisiana, May 2026.

3. *Complete Decentralization of Linear Quadratic Gaussian Control for the Discrete Wave Equation*, IEEE 64th Conference on Decision and Control, Rio de Janeiro, Brazil, December 2025.

2. *Deconvolution of Temporally Under-Resolved Image Sequences for Coupled Dynamical Systems*, Joint Mathematics Meeting, (virtual), January 2022.

1. *Exploring the garden of Petaluma knots*, Nebraska Conference for Undergraduate Women in Mathematics, Lincoln, Nebraska, January 2020.

Teaching and Mentorship Experience

Discovery Learning Apprenticeship (DLA) Mentor (Aug 2026 – May 2027)
University of Colorado Boulder
Primary advisor to undergraduate student Elena Abbruzzi in university funded research.

Graduate Teaching Assistant (Aug 2022 – Dec 2023)
University of Colorado Boulder
Taught recitation and workgroup sections, held office hours, and wrote and graded quizzes, homeworks, and projects for courses such as Calculus 1 and 2 for Engineers and Introduction to Differential Equations with Linear Algebra.

Math Tutor (Feb 2020 – May 2022)
University of St. Thomas - Minnesota
Peer tutor for all math courses through multivariable calculus. Included training in diversity, equity, and inclusion and tutoring best practices

Service

Association for Women in Mathematics

President: Boulder AWM student chapter (May 2024 – Apr 2026)
Treasurer: Boulder AWM student chapter (May 2023 – Apr 2024)
Member: National AWM organization and Boulder AWM student chapter (Aug 2022 – present)

Reviewer

Conference on Decision and Control

Volunteer

American Control Conference - Denver, Colorado (July 2025)